# Topic quiz: Power and efficiency

IBDP Physics SL · Grade 11 · A.3 Power and efficiency · 10 multiple-choice questions

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Name: Class: Date:

**How to use.** Practise with the simulations and worksheets first. Then answer every question: circle one letter and write it in the grid below. Use g = 9.81 m $s^{-2}$ and show any working next to the question.
**Check yourself online:** the same quiz with instant feedback and explanations — https://school.taneski.com/physics/dp/quiz-power-efficiency/

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## Question 1

A 60.0 kg student runs up stairs with a vertical height of 3.0 m in 4.0 s. What is the useful power output?

(A) 45.0 W
(B) 441 W
(C) 1770 W
(D) 7060 W

## Question 2

A car travels at a constant 25 m $s^{-1}$ against a total resistive force of 800 N. What power does the engine deliver to the wheels?

(A) 0 W, because the resultant force is zero
(B) 32 W
(C) 800 W
(D) 20 kW

## Question 3

A motor raises an 800 kg lift at a constant speed of 1.5 m $s^{-1}$. What is the useful power output of the motor?

(A) 0.90 kW
(B) 1.2 kW
(C) 5.2 kW
(D) 11.8 kW

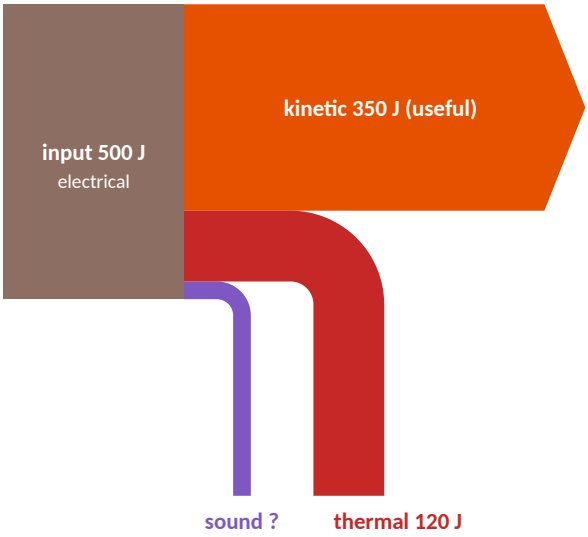
## Question 4

An electric motor uses 3.5 kJ of energy to lift a 150 N load through 7.2 m. What is its efficiency?

(A) 3.1 %
(B) 31 %
(C) 69 %
(D) 324 %

### Question 5

The Sankey diagram shows the energy transfers in an electric motor every second. What is the missing sound energy and the efficiency of the motor?




(A) Sound 30 J, efficiency 70 %
(B) Sound 30 J, efficiency 30 %
(C) Sound 150 J, efficiency 70 %
(D) Sound 30 J, efficiency 74 %

### Question 6

A ball is dropped from 1.20 m and bounces back up to 0.84 m. What percentage of its energy does it keep after the bounce?

(A) 30 %
(B) 70 %
(C) 84 %
(D) 143 %

### Question 7

Petrol has an energy density of 46 MJ $kg^{-1}$. A car carries 40 kg of petrol, its engine is 25 % efficient, and at 25 m $s^{-1}$ it needs 20 kW of useful power. About how far can it travel on this petrol?

(A) 23 km
(B) 144 km
(C) 575 km
(D) 2300 km

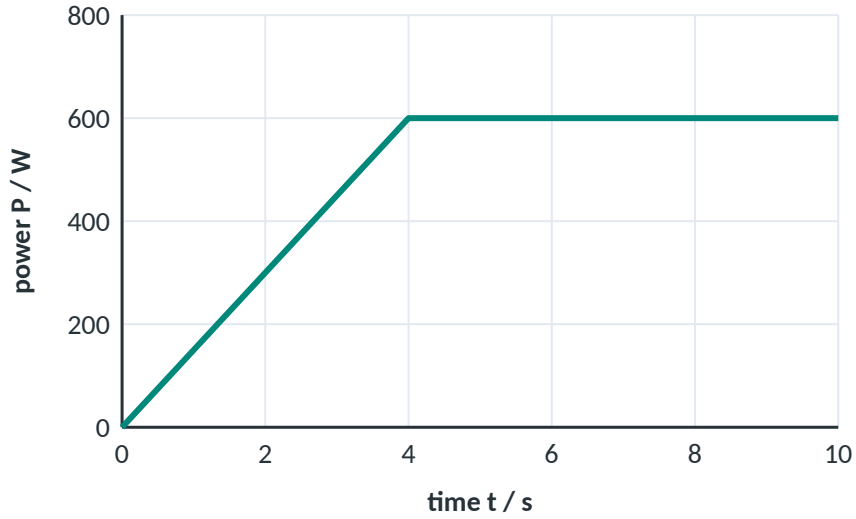
### Question 8

Why can no machine have an efficiency greater than 100 %?

(A) The useful energy out cannot be more than the total energy in, because energy is conserved
(B) Machines always need some input power
(C) Friction always wastes exactly half the energy
(D) Power is always lost faster than energy

## Question 9

The graph shows the power output of a motor. How much energy does it transfer in the 10 s?




(A) 60 J

(B) 3600 J

(C) 4800 J

(D) 6000 J

## Question 10

A 1200 kg car drives up a 5.0° slope at a constant 15 m $s^{-1}$ against a resistive force of 600 N. What is the power output of the engine?

(A) 9.0 kW

(B) 15.4 kW

(C) 24.4 kW

(D) 185 kW

End of quiz · Score: _____ / 10